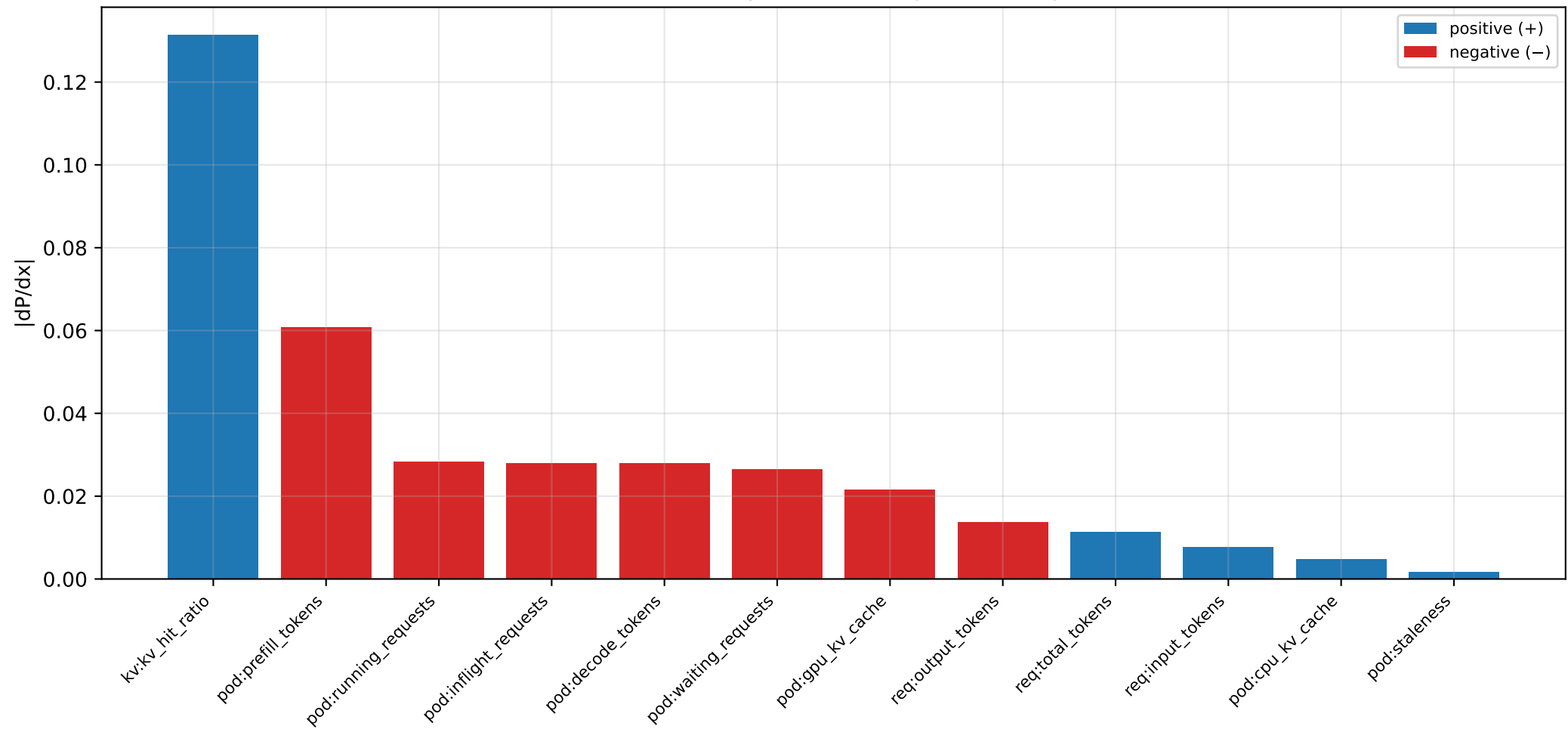


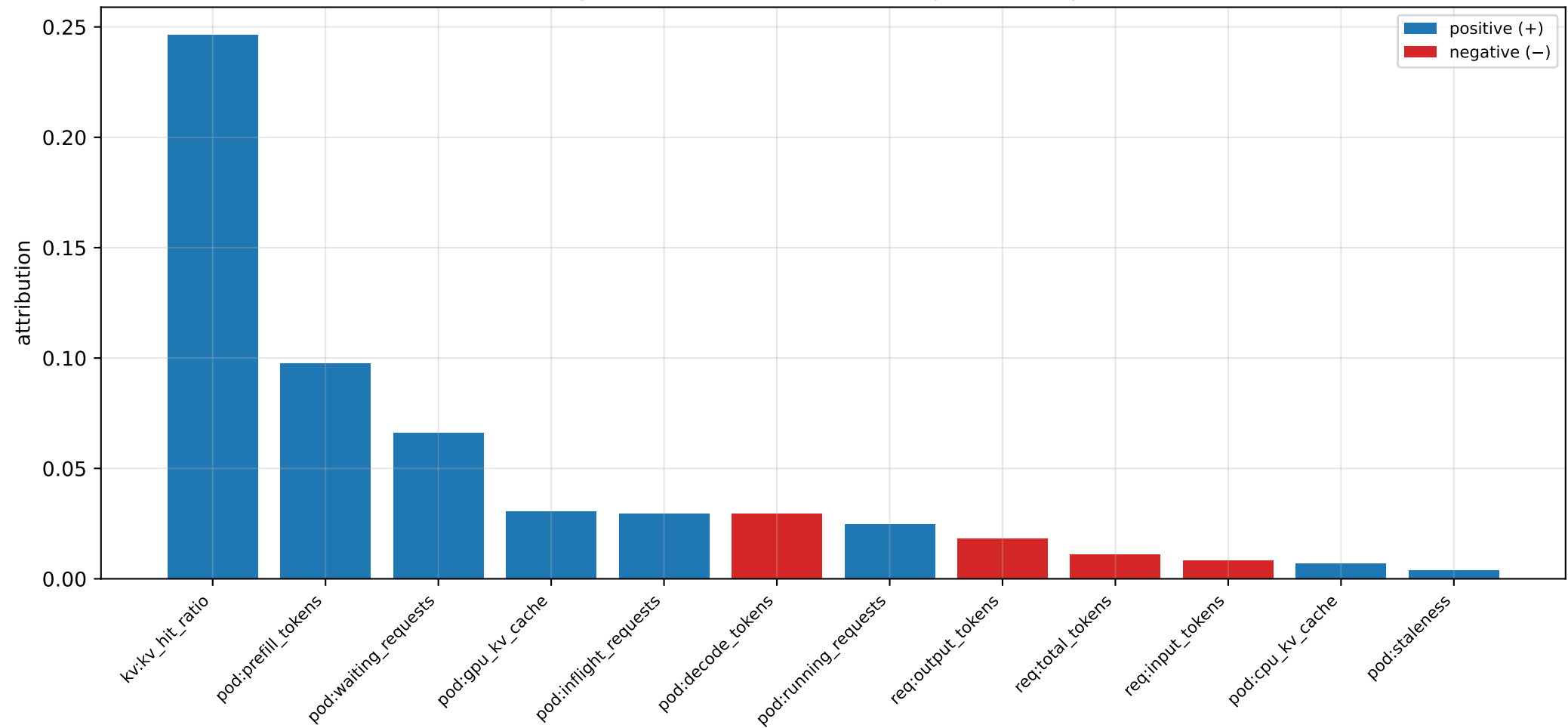
Routing Policy Explainability Report

final_model-data_replaced-processed-linear_simple-lr_0.001-ttft_weight_2.0-ttftslo_1000-avgtpotslo_50

Gradient Saliency (combined: pod, kv, request)



Integrated Gradients (combined: pod, kv, request)



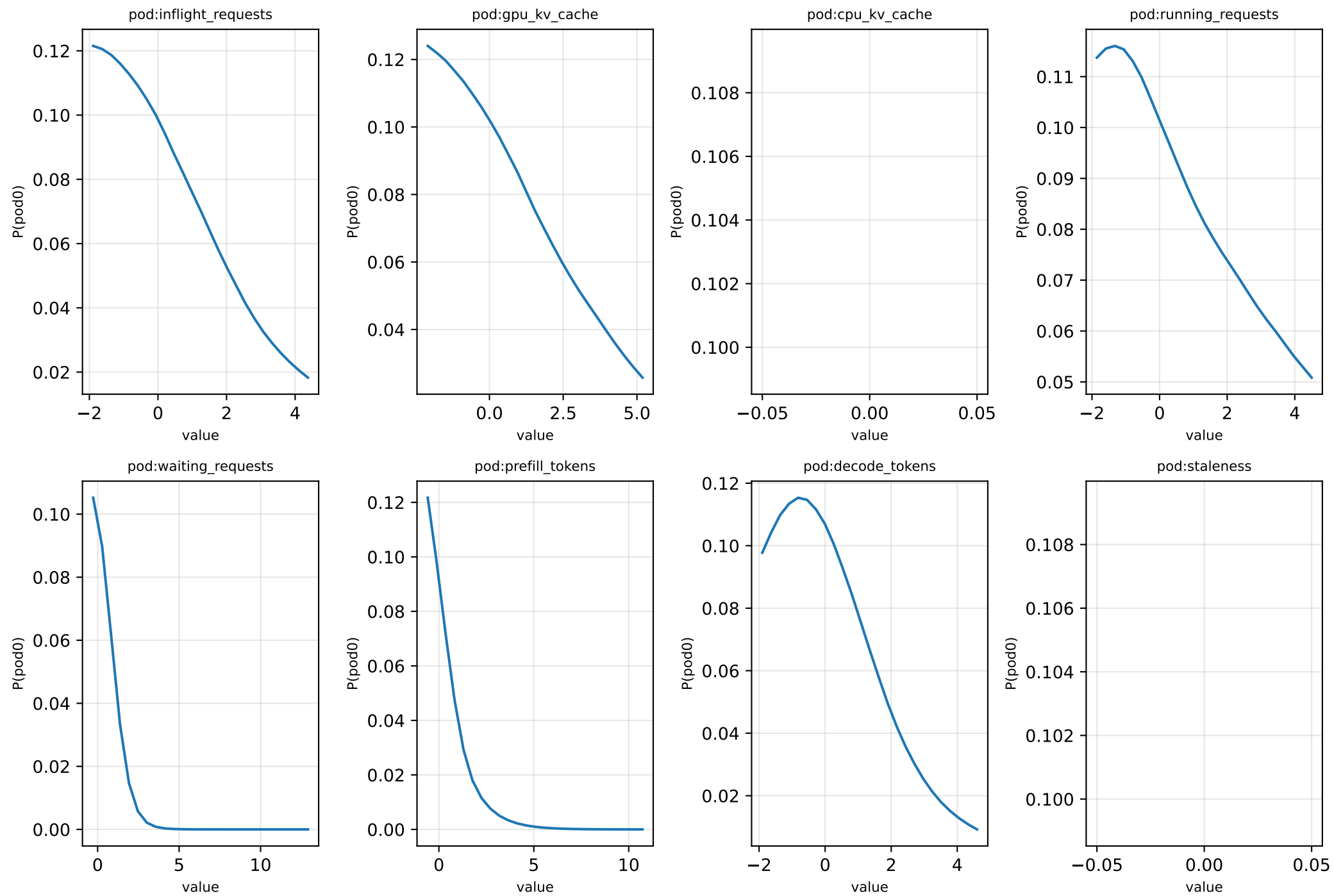
Permutation Symmetry Test

match_rate: 1.0

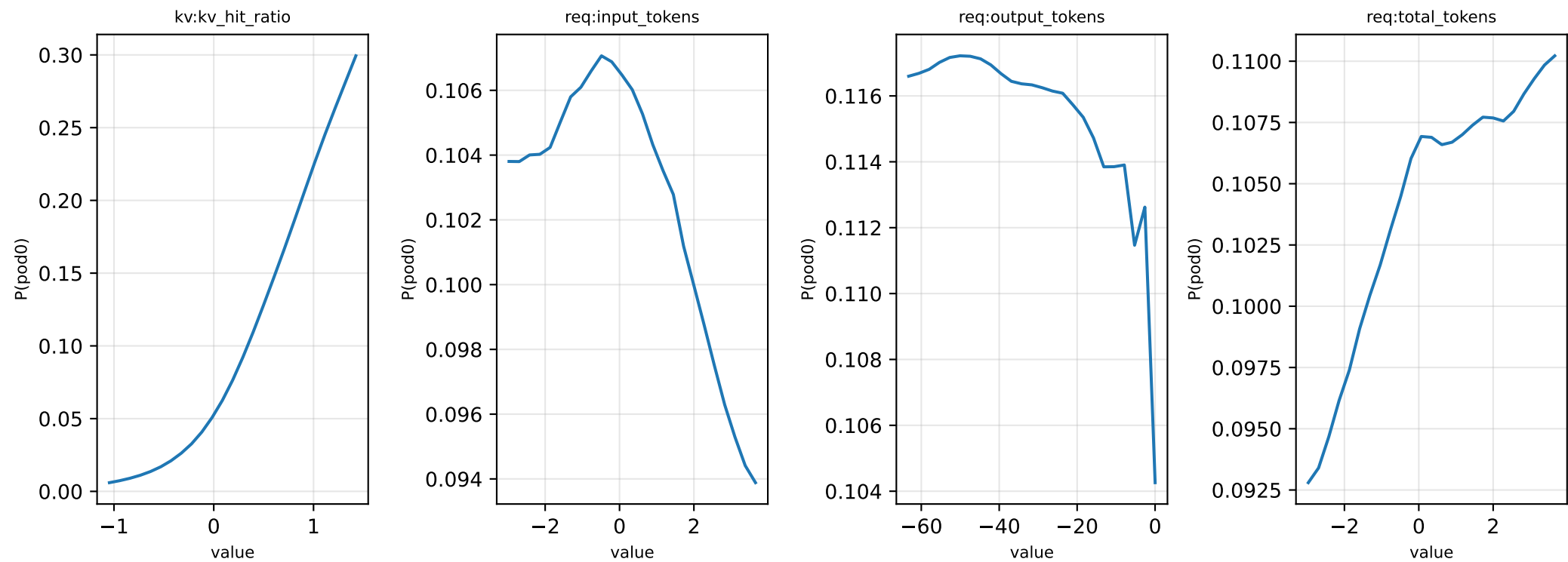
mean_l1_diff: 1.3588439769692447e-08

Higher match_rate and lower mean_l1_diff indicate invariance to pod ordering.

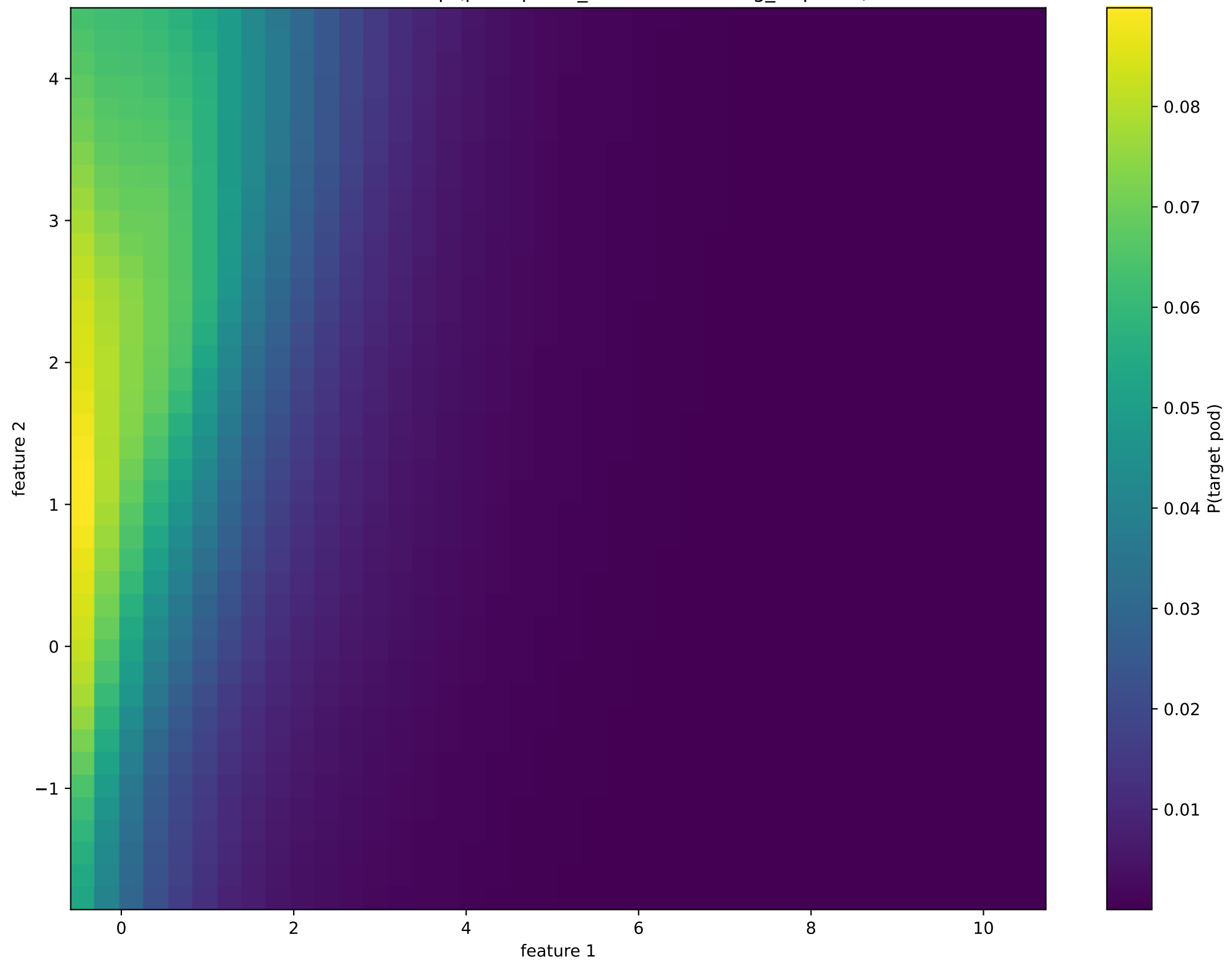
PDPs (all groups), target pod=0



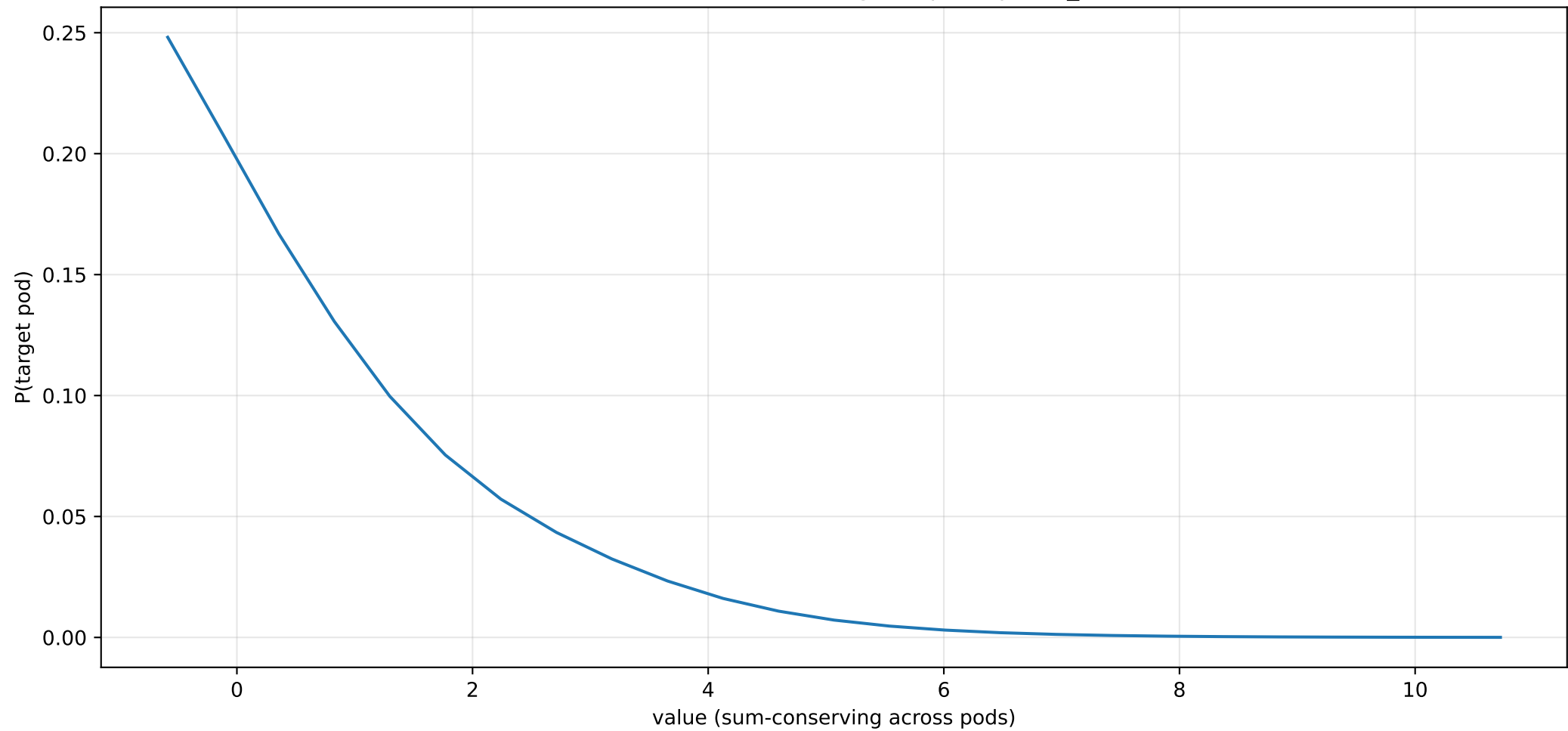
PDPs (all groups), target pod=0



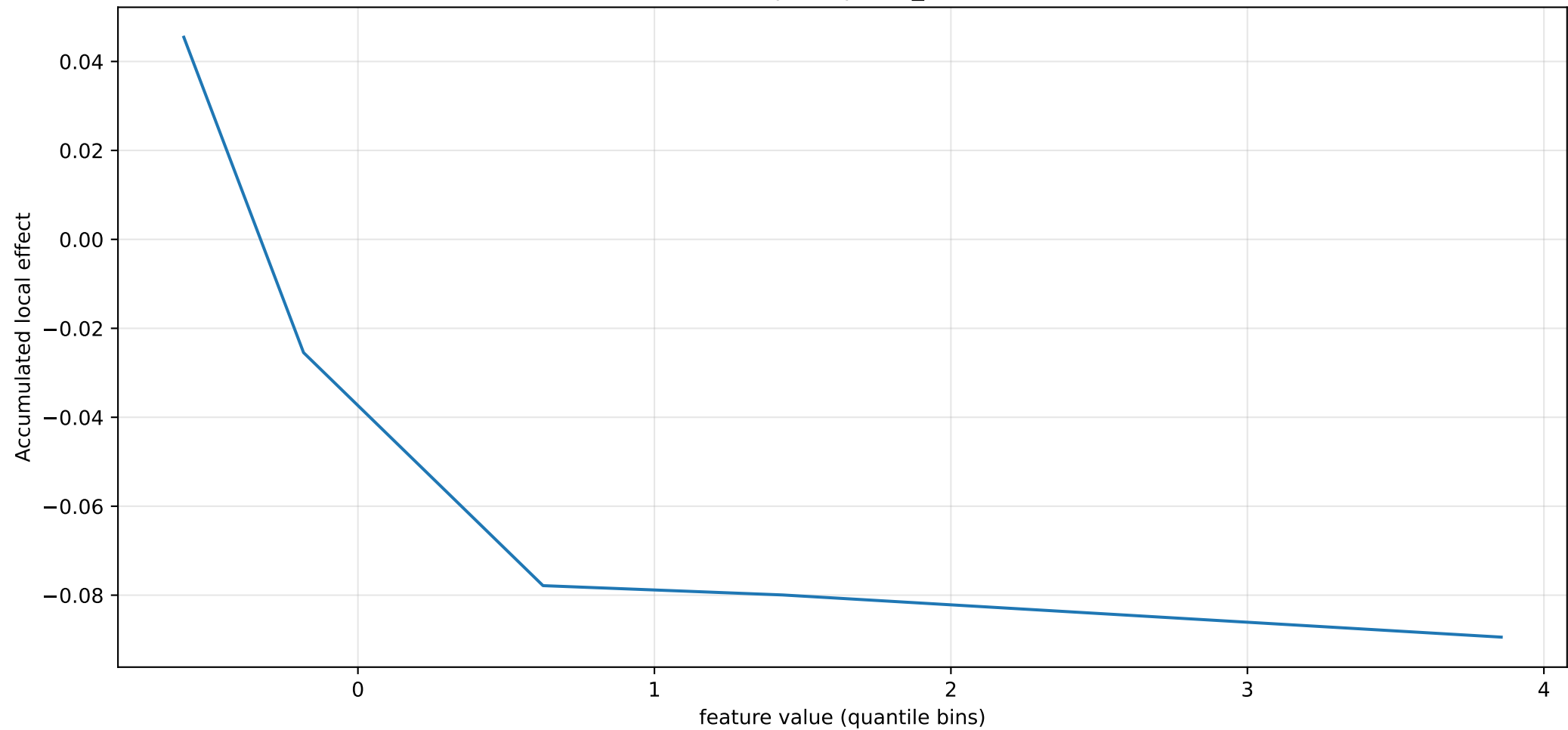
Interaction heatmap (pod0 prefill_tokens vs running_requests)



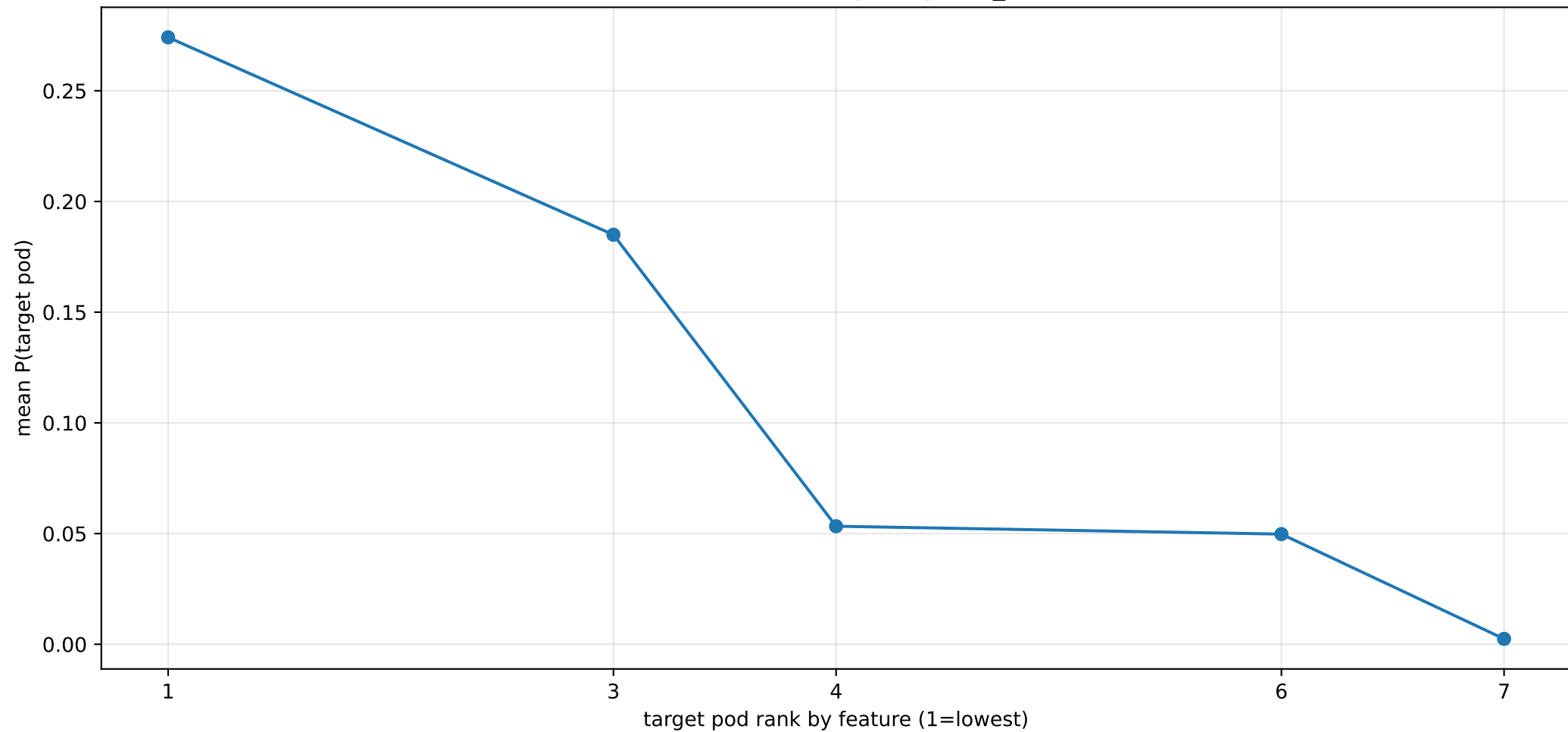
Relative PDP (sum-conserving) for pod0:prefill_tokens



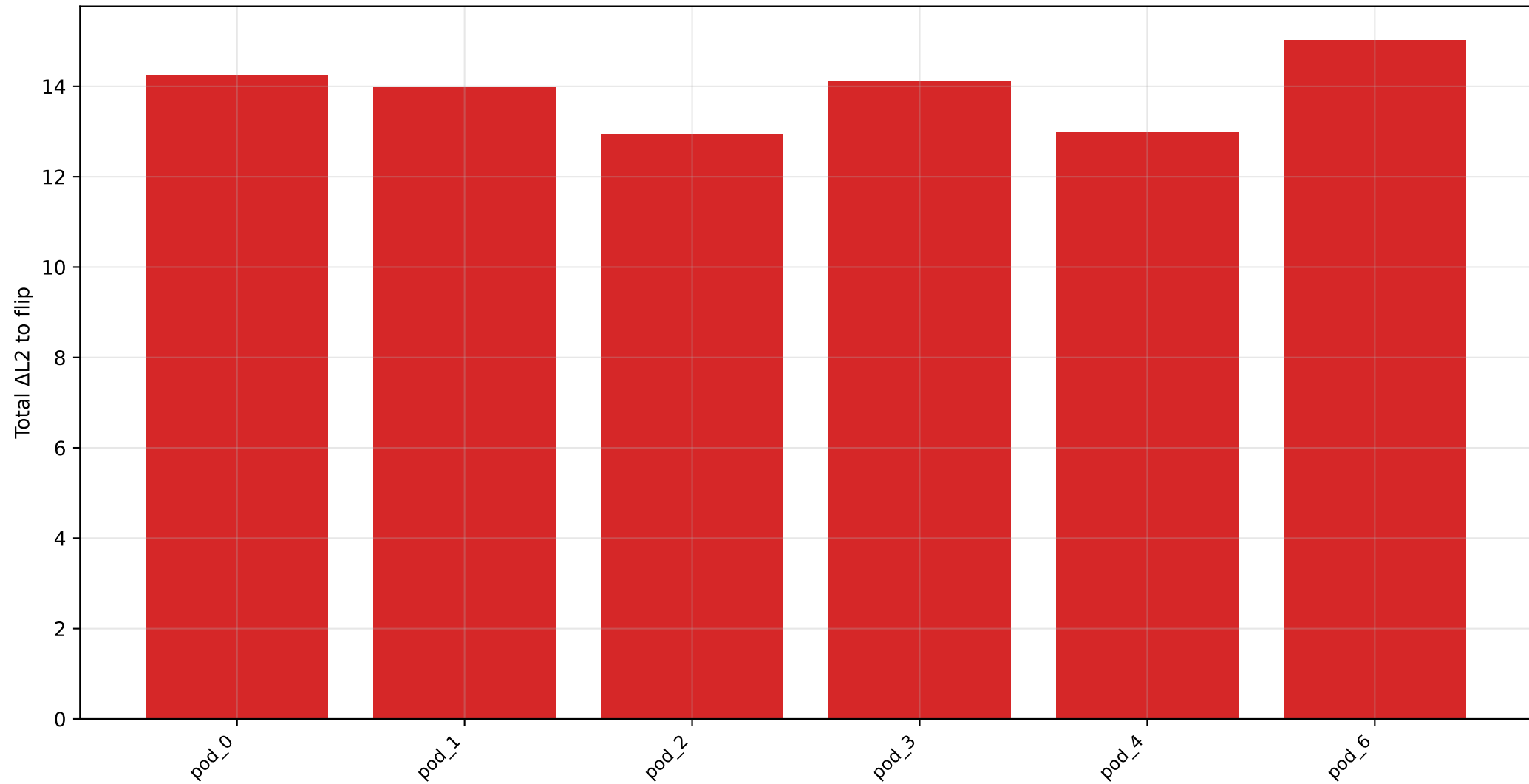
ALE for pod0:prefill_tokens



Rank-based effect for pod0:prefill_tokens



Counterfactual Distance to Flip to Each Pod (green=success, red=failed within budget)



EXPLAINABILITY FINDINGS

Saliency (local sensitivity):

- prefill_tokens: 0.061
- running_requests: 0.028
- inflight_requests: 0.028
- decode_tokens: 0.028
- waiting_requests: 0.026
- gpu_kv_cache: 0.022
- cpu_kv_cache: 0.005
- staleness: 0.002

Request Saliency:

- output_tokens: 0.014
- total_tokens: 0.011
- input_tokens: 0.008

KV Saliency:

- kv_hit_ratio: 0.132

Integrated Gradients (robust attribution):

- Top pod feature: prefill_tokens
- KV importance: 0.247
- Request max attribution: output_tokens

PDP Trend Summary (target pod)

Pod features:

- inflight_requests: decreasing
- gpu_kv_cache: decreasing
- cpu_kv_cache: flat
- running_requests: decreasing
- waiting_requests: decreasing
- prefill_tokens: decreasing
- decode_tokens: decreasing
- staleness: flat

KV features:

- kv_hit_ratio: increasing

Request features:

- input_tokens: decreasing
- output_tokens: decreasing
- total_tokens: increasing

Permutation Symmetry:

- match_rate: 1.0
- mean_l1_diff: 1.3588439769692447e-08

Counterfactual Robustness:

- success_rate (flip achieved): 0.00
- min $\Delta L2$ to flip: 12.95
- mean $\Delta L2$ to flip: 13.88

Interpretation:

- High symmetry indicates no position bias.
- Larger $\Delta L2$ implies locally robust decisions (harder to flip).